

Could not connect to the reCAPTCHA service. Please check your internet connection and reload to get a reCAPTCHA challenge.

```
import pandas as pd

data = {
    'name': ['Alice', 'Bob', 'Charlie', 'David', 'Emma', 'Frank', 'Grace',
            'Henry', 'Isabella', 'Jack', 'Karen', 'Liam', 'Mia', 'Noah', 'Olivia'],
    'gender': ['Female', 'Male', 'Male', 'Male', 'Female', 'Male', 'Female',
              'Male', 'Female', 'Male', 'Female', 'Male', 'Female', 'Male', 'Female'],
    'age': [25, 30, 28, 35, 27, 40, 32, 29, 26, 31, 33, 24, 36, 38, 29],
    'department': ['HR', 'IT', 'Finance', 'IT', 'HR', 'Sales', 'Finance',
                  'IT', 'Sales', 'HR', 'Finance', 'IT', 'Sales', 'Finance', 'HR'],
    'performance_score': [650, 720, 580, 810, 490, 670, 730, 550, 760, 620, 690, 450, 850, 710, 540]
}

df = pd.DataFrame(data)

df.to_csv('employee.csv', index=False)

print(df.head())
```

	name	gender	age	department	performance_score
0	Alice	Female	25	HR	650
1	Bob	Male	30	IT	720
2	Charlie	Male	28	Finance	580
3	David	Male	35	IT	810
4	Emma	Female	27	HR	490

```
df = pd.read_csv('employee.csv')
```

```
import pandas as pd

df = pd.read_csv('employee.csv')

print(df.head())
print(df.info())
print(df.describe())
print("Shape:", df.shape)
```

	name	gender	age	department	performance_score
0	Alice	Female	25	HR	650
1	Bob	Male	30	IT	720
2	Charlie	Male	28	Finance	580
3	David	Male	35	IT	810
4	Emma	Female	27	HR	490

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 15 entries, 0 to 14
Data columns (total 5 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   name            15 non-null    object
1   gender          15 non-null    object
2   age             15 non-null    int64
3   department      15 non-null    object
4   performance_score 15 non-null    int64
dtypes: int64(2), object(3)
memory usage: 732.0+ bytes
None
```

	age	performance_score
count	15.000000	15.000000
mean	30.866667	654.666667
std	4.778922	115.811834
min	24.000000	450.000000
25%	27.500000	565.000000
50%	30.000000	670.000000
75%	34.000000	725.000000
max	40.000000	850.000000

```
Shape: (15, 5)
```

```
df.set_index('name', inplace=True)

print(df.head())
```

	gender	age	department	performance_score
Alice	Female	25	HR	650
Bob	Male	30	IT	720
Charlie	Male	28	Finance	580
David	Male	35	IT	810
Emma	Female	27	HR	490

```
high_perf = df[df['performance_score'] > 700]
```

```
print(high_perf)
```

name	gender	age	department	performance_score
Bob	Male	30	IT	720
David	Male	35	IT	810
Grace	Female	32	Finance	730
Isabella	Female	26	Sales	760
Mia	Female	36	Sales	850
Noah	Male	38	Finance	710

```
mid_perf = df[
    (df['performance_score'] > 500) &
    (df['performance_score'] < 700)
]

print(mid_perf)
```

name	gender	age	department	performance_score
Alice	Female	25	HR	650
Charlie	Male	28	Finance	580
Frank	Male	40	Sales	670
Henry	Male	29	IT	550
Jack	Male	31	HR	620
Karen	Female	33	Finance	690
Olivia	Female	29	HR	540

```
print(df.isnull().sum())
```

```
gender          0
age             0
department      0
performance_score 0
dtype: int64
```

```
df.fillna(df.select_dtypes(include='number').median(), inplace=True)
```

```
lower = df['performance_score'].quantile(0.05)
upper = df['performance_score'].quantile(0.95)

print("Lower limit:", lower)
print("Upper limit:", upper)

df['performance_score'] = df['performance_score'].clip(lower, upper)
```

```
Lower limit: 478.0
Upper limit: 822.0
```

```
outliers = df[
    (df['performance_score'] < lower) |
    (df['performance_score'] > upper)
]

print(outliers)
```

```
Empty DataFrame
Columns: [gender, age, department, performance_score]
Index: []
```

```
df_encoded = pd.get_dummies(
    df,
    columns=['gender'],
    drop_first=True
)

print(df_encoded.head())
```

name	age	department	performance_score	gender_Male
Alice	25	HR	650	False
Bob	30	IT	720	True
Charlie	28	Finance	580	True
David	35	IT	810	True
Emma	27	HR	490	False

```
print(df_encoded.info())
print(df_encoded.head())
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 15 entries, Alice to Olivia
Data columns (total 4 columns):
```

```
#   Column      Non-Null Count  Dtype
---  -
0   age         15 non-null    int64
1   department   15 non-null    object
2   performance_score 15 non-null    int64
3   gender_Male  15 non-null    bool
dtypes: bool(1), int64(2), object(1)
memory usage: 1.0+ KB
None
```

	age	department	performance_score	gender_Male
name				
Alice	25	HR	650	False
Bob	30	IT	720	True
Charlie	28	Finance	580	True
David	35	IT	810	True
Emma	27	HR	490	False